The System of Context

The Ecosystem, Chapter 6/9 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/ecosystem-system-of-context

EN: A System of Record decides what is TRUE. A System of Context decides what ARRIVES. It's the connecting layer between a vertical SoR and a customer's real applications/identities -- it carries authority, but never holds it.
Roman Urdu: System of Record decide karta hai kya TRUE hai. System of Context decide karta hai kya ARRIVE hota hai. Yeh vertical SoR aur customer ke real applications/identities ke beech connecting layer hai -- authority carry karta hai, hold nahi karta.

1. Three Layers, Three Jobs

- Systems of Record (what's officially true) -> System of Context (what's relevant now) -> Worker (reason + act).
- **One Law:** the System of Context carries authority, it never holds it. A model is never a source -- no register row, plausibility isn't provenance, it can't be a permission-enforcement point.
- 3 quiet leaks to watch for: gap-filling, drifting paraphrase, unmanaged persistent model memory.

2. Routing + the Context Packet

- Authority is federated -- many records, each with its own authority, never merged. 3 paths: rule (governed retrieval), number (live typed query), case-talk (ungoverned, supporting context only).
- 7 operations: connect, identify, retrieve, filter, rank, assemble, deliver. The Context Packet has 8 compartments, is temporary, inherits authority+permissions, and expires.

3. Permission, Provenance, Conflict, 8 Invariants

- Permission must be enforced BEFORE the model sees the data, not after. Provenance = 9 fields tracked per item. Conflicts resolve by scope, by authority, or escalate.
- **8 Invariants:** authority, relevance, permission, freshness, provenance, conflict, promotion, discovery-vs-confirmation.

4. Failure Modes + Templates

- 11 named failure modes include: flat index, reliance on the copy, stale answer, model-as-quiet-source, citations without provenance, shadow record.
- 4 templates: connected source register, authority routing map, boundary contract, permission test set (must include a 'nothing' test).

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
System of Context	The layer that decides what arrives, carrying but never holding authority.	Layer jo decide karta kya arrive hota hai, authority carry karta par hold nahi karta.
Context Packet	A temporary, 8-compartment bundle that inherits authority/permissions and expires.	Temporary, 8-compartment bundle jo authority/permissions inherit karta aur expire hota hai.
Fail closed (permission)	Permission enforcement happens before the model sees data, not after.	Permission enforcement model ko data dikhane se pehle hoti hai.
Provenance	9 tracked fields recording where each piece of context came from.	9 fields jo track karte context kahan se aaya.
8 Invariants	Authority, relevance, permission, freshness, provenance, conflict, promotion, discovery-vs-confirmation.	Authority, relevance, permission, freshness, provenance, conflict, promotion, discovery-vs-confirmation.

Quick Self-Check

Question	Answer
What's the core distinction between a System of Record and a System of Context?	SoR decides what is true; System of Context decides what arrives (relevant now) -- it carries authority but never holds it.
Why can't a model itself be a 'source'?	No register row, plausibility isn't provenance, it can't enforce permissions -- it can only repeat or paraphrase.
What are the 3 quiet leaks in context delivery?	Gap-filling, drifting paraphrase, and unmanaged persistent model memory.
Why must a permission test set include a 'nothing' test?	To prove the system can correctly return NO results for someone without access, not just correctly filter for someone with it.

Where This Leads / Sources

Leads to: Building the Context Layer (hands-on build), Designing the Vertical System of Record, AI Identity crash course.

Source: docs/ecosystem-system-of-context/ (this repo), content originally via Zia Tutor AI connector.